MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSEECT200 ARM based Embedded System

Duration: 2 hours **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	CISC architecture is characterized by a) Fixed-length instructions b) Reduced number of instructions c) Variable-length instructions d) Load-store architecture only	K	1
2	Which ARM processor family is widely used in low-power embedded applications? a) ARM10 b) ARM11 c) ARM7 d) ARM Cortex-A		2
3	ARM processors typically use a) Stack-based architecture b) Accumulator-based architecture c) Load–store architecture d) Memory–memory architecture	U	3
4	ARM logical instructions include a) ADD, SUB b) AND, ORR, EOR c) LDR, STR d) B, BL	K	1
5	Which program development tool executes code without real hardware? a) Emulator b) Simulator c) IDE d) Compiler	K	1
6	Which C data type occupies the least memory? a) int b) float c) char d) double	K	1
7	Register allocation in Embedded C refers to a) Assigning registers to peripherals b) Mapping variables to CPU registers c) Allocating memory in heap d) Allocating I/O ports	U	3
8	The instruction used to call a function in ARM assembly is a) B b) BL c) BX d) SWI	K	1
9	Which operating mode is used during normal program execution in ARM? a) IRQ mode b) FIQ mode c) Supervisor mode d) Thread mode	K	2
10	Which memory region is typically used for program code in Cortex-M? a) SRAM b) Peripheral region c) Flash memory d) External DRAM	K	2

11	Which register acts as the program counter? a) R13 b) R14 c) R15 d) APSR	K	2
12	Which peripheral is commonly used for digital I/O? a) ADC b) DAC c) GPIO d) PWM	K	3
13	UART communication is a) Synchronous b) Asynchronous c) Half-duplex only d) Multi-master	K	3
14	A switch connected to a microcontroller input usually requires a) Pull-up or pull-down resistor b) Crystal oscillator c) ADC d) Op-amp	U	4
15	LM35 temperature sensor output is proportional to a) Current b) Resistance c) Temperature in °C d) Light intensity	K	4
16	The AMBA bus architecture is mainly used for a) External networking b) Peripheral communication c) Memory management d) Power management	U	2

1 * 16 = 16

Part B

Short Answer Questions
Answer any **6** Questions
Each question carries 5 marks

17	Compare Harvard vs Von Neumann architecture	U	1
18	What is the difference between IRQ and FIQ exceptions?	K	2
19	What is GPIO?	U	3
20	What is meant by Thumb instruction set	U	1
21	Differentiate between Thread mode and Handler mode.	U	2
22	Which ARM GPIO pin modes are used for interfacing an IR sensor?	K	4
23	What is I²C communication?	K	3
24	Explain the sequence of operation of a traffic light controller.	U	4

6 * 5 = 30

Part C

Essay Questions
Answer any **2** Questions
Each question carries 12 marks

25	Explain embedded systems in detail. Discuss their characteristics, components, and applications.	K	1
26	Discuss the register organization in ARM Cortex-M processors. Explain the role of special-purpose registers.	U	2
27	Describe GPIO interfacing in embedded systems. Explain LED toggling, pin masking and ON–OFF control	U	3
28	Write an embedded C program to control a relay using a push button	U	4

2 * 12 = 24